25900407 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEFSC200 - INSTRUMENTATION - CHEMICAL

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer in one word

Answer all questions

Each question carries **1** marks

1. What are molecular energy levels? [U] / [CO1]
2. The first electrode struck by ions in an Electron Multiplier is called the ___. [A] / [CO2]
3. Name one common detector used in HPLC. [U] / [CO3]
4. Mention one type of injection system used in GC. [K] / [CO4]
5. Name one type of dispersive IR instrument? [K] / [CO1]
6. State Lambert's law. [K] / [CO1]
7. State the basic principle of LC-MS. [U] / [CO3]
8. Name any one type of elution technique in HPLC ? [U] / [CO3]
9. Mention one forensic application of GC. [K] / [CO4]

10. MS aids in monitoring ___ levels in environmental samples. [A] / [CO2]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

11. Write the functions of source, monnochromator and detector. [U] / [CO1]

12. Examine the working principle of Tandem Mass Spectrometry (MS/MS) and its analytical applications. [A] / [CO2]

13. Mention three common solvents used in TLC. [U] / [CO3]

14. What is the basic principle of GC-MS? [U] / [CO4]

15. What is the role of a detector in IR spectroscopy? [U] / [CO1]

16. Explain Jablonski diagram briefly. [U] / [CO1]

17. Determine the principle of Mass Spectrometry (MS). [A] / [CO2]

[3x5 = 15]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Discuss the contribution of detectors in improving chromatographic analysis in forensic investigations. [An] / [CO3]

19. Compare IR, FTIRand Raman spectroscopy? [U] / [CO1]

20. Explain the width of absorption lines and their role in interpretation of NMR? [U] / [CO2]

21. Discuss energy transitions in UV - visible spectroscopy. [U] / [CO1]

22. Define the different flame fuel/oxidant combinations used in flame-based ionization techniques and explain how they affect [A] / [CO2]

23. Explain capacity factor, selectivity factor, and column resolution with formulas. [An] / [CO3]

24. Explain the spin properties of nuclei in NMR? [U] / [CO2]

[6x5 = 30]

Part D

Essay Type Questions

Answer any **1** questions

Each question carries **15** marks

25. Explain the interaction of electromagnetic radiation with matter in detail with examples.	[U] / [CO1]
26. Define Scanning modes SIM & Full SCAN	[A] / [CO2]

[15x1 = 15]